

Introduction to model-based community analysis

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Gathering data

We go out, register species at multiple sites



Figure 1: Geir-Harald Strand / NIBIO

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What does community data look like?

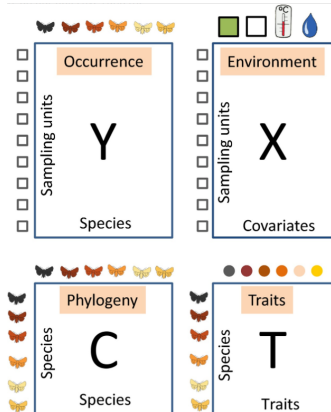
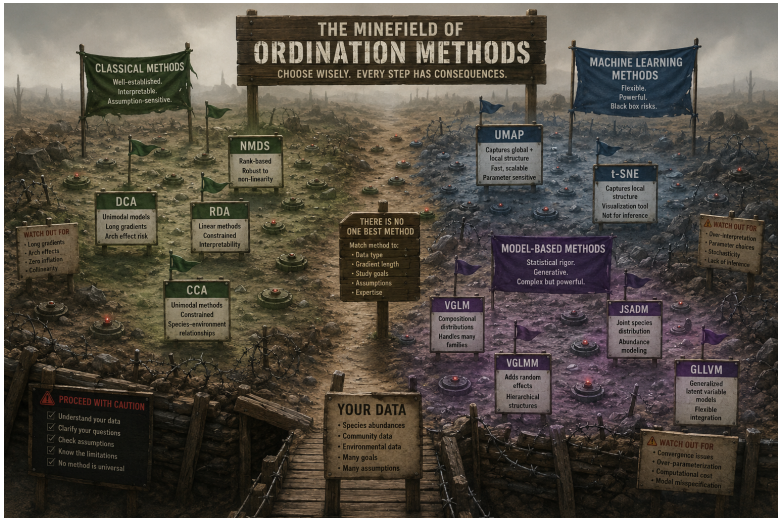


Figure 2: Ovaskainen et al. (2017) Fig. 3

Now pick an analysis



Outline of the workshop

Day	Topics
1	Vector GLMs → Vector GLMMs → Model checking
2	Hierarchical responses → JSDMs → Diversity
3	Model-based ordination → Constrained/concurrent → Conditioning
4	Unimodal responses → Spatial/temporal → Own data

This lecture: why model-based methods, properties of community data, and how models address them.

From VGLMs and JSDBMs to ordination and beyond

- ▶ **VGLMs:** multispecies regression; species-specific responses to the environment
- ▶ **JSDBMs:** add residual species correlations; why do species co-occur after accounting for the environment?
- ▶ **Ordination:** latent variables become the primary object; what are the main unmeasured gradients driving the community?

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Each step builds on the previous. All use the same statistical framework (GLLVMs).

Community ecology

We collect data on many species at many sites. We want to understand:

- ▶ Which species co-occur, and why?
- ▶ How do species respond to the environment?
- ▶ How do communities change across gradients?
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They have serious limitations.

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When they fail: long gradients, sparse count data, hierarchical study designs, prediction under new conditions.

This workshop teaches the framework that makes the approximation explicit and testable.

`gllvm`, `Hmsc`, and `mvabund` are now standard tools in the community ecology literature.

A strategy by necessity

Community ecology needed tools for multispecies data long before the statistical machinery existed to model it properly.

- ▶ Distance matrices and transformations were the best available options in the 1970s–90s
- ▶ They became standard practice: embedded in textbooks, software, and reviewer expectations
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Now there is.

What do classical methods assume?

Method	Purpose	Assumptions
PCA	Unconstrained	Gaussian; linear; constant variance
RDA	Constrained	Gaussian; linear; constant variance
CA	Unconstrained	Approximate species packing
CCA	Constrained	Approximate species packing
DCA	Unconstrained	Approximate species packing; rescaled
NMDS	Unconstrained	Distance metric reflects ecological similarity
db-RDA	Constrained visualisation	Distance metric reflects ecological similarity
PERMANOVA	Hypothesis test	Homogeneous within-group dispersion

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A single process-oriented framework can address all of these tasks.
 We will return to this throughout the workshop.

What do classical methods do well?

These methods have been the backbone of community ecology for 50+ years, and for good reason:

- ▶ **Easy to use:** `vegan` is well-documented; functions are fast and accessible
- ▶ **Known behavior:** artifacts and failure modes are well-characterised; large published literature
- ▶ **Computationally fast:** eigenvalue decomposition scales well; permutation is parallelisable
- ▶ **Permutation inference:** PERMANOVA and ANOSIM require no distributional assumptions for the test statistic
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Pattern recognition cannot answer process questions. A model for the data-generating process can.

Classical methods: no random effects

Classical ordination cannot accommodate hierarchical study designs:

- ▶ Repeated measurements of sites
- ▶ Plots nested in sites
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Adding a random effect to CCA or NMDS is not possible.

In model-based analysis, mixed effects are a natural extension of the model, the same tools you use in a GLMM.

Classical methods: no uncertainty

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Model-based methods inherit full uncertainty quantification from the statistical model.

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Model-based methods define a data-generating process.

Predictions for new sites, new conditions, and derived quantities follow from the model.

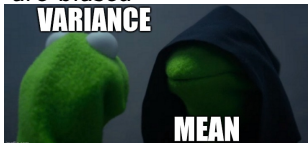
Classical methods: transformation or distances

Community data is sparse, overdispersed, and non-normal.

Classical methods apply one of two workarounds:

Data transformation

- ▶ $\log(y + 1)$, square root, presence/absence
- ▶ Does not remove the mean-variance trend (O'Hara & Kotze 2010)
- ▶ Variance not stabilized; back-transformed estimates are biased



Distance matrix

- ▶ Bray-Curtis, Jaccard, Euclidean, ...
- ▶ Discards raw data structure
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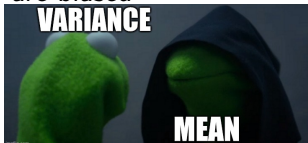
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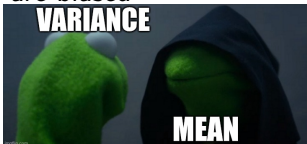
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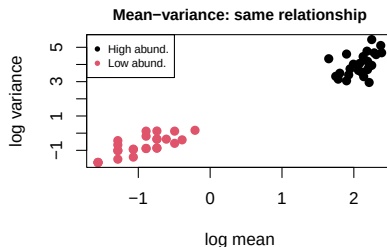
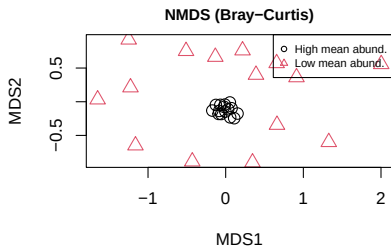


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Distance-based methods confound location and dispersion

Warton et al. (2012): when communities differ only in mean abundance (a location effect), Bray-Curtis ordination suggests a dispersion effect.



Low-abundance group appears more dispersed in NMDS. The mean-variance plot shows both groups share the same relationship. **There is no dispersion effect.** The apparent dispersion is an artefact of Bray-Curtis weighting.

The model-based alternative

Two papers that directly motivated this workshop:



Forum Open Access

The central role of mean-variance relationships in the analysis of multivariate abundance data: a response to Roberts (2017)

David I. Warton Francis K. C. Hui

First published: 26 July 2017 | <https://doi.org/10.1111/2041-210X.12843> | Citations: 28

Warton & Hui (2017): mean-variance relationships are
central; distance-based analyses get them wrong

Plant Ecol (2015) 216:669–682
DOI 10.1007/s11258-014-0366-3

Model-based thinking for community ecology

David I. Warton · Scott D. Foster ·
Glenn De'ath · Jakub Stoklosa · Piers K. Dunstan

Warton et al. (2015): a framework for model-based
thinking in community ecology

Contemporary model-based methods

In classical ecology, ordination *is* the analysis. In the GLLVM framework, ordination is the tool of choice when it is not possible to estimate species-specific parameters.

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- ▶ **JSDMs**: incorporate species correlations via latent variables
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We begin with full-rank models on Days 1–2 because they make fewer assumptions. Ordination is introduced on Day 3, once the foundations are in place.

All three can be fitted with the `gllvm` R package.

Models are not a panacea

- ▶ **Asymptotic results:** inference relies on large-sample approximations; standard errors and p-values can be unreliable for small datasets or sparse community matrices
- ▶ **Likelihood approximations:** `gllvm` uses a variational approximation; it is fast but not exact, and quality varies with model complexity and data sparsity
- ▶ **Computationally intensive:** complex models with many species and latent variables are slow; Bayesian alternatives (e.g. `Hmsc`) are slower still
- ▶ **Steep learning curve:** model specification, convergence diagnosis, and interpretation require more statistical knowledge than running `vegan`

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The goal is not to replace judgment with software, but to make

Two processes, one model

Our data results from two distinct processes:

The ecological process

- ▶ Species-environment relationships
- ▶ Co-occurrence and interactions
- ▶ Diversity and community assembly

Our primary interest.

The sampling process

- ▶ Hierarchical study design
- ▶ Preferential or opportunistic sampling
- ▶ Imperfect detection
A source of noise.

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Classical methods: sampling accommodated through study design alone.

Model-based: random effects for hierarchical designs, offsets for effort, detection submodels; the model reflects how the data was collected.

Joint Species Abundance Distribution Models

After accounting for the environment, species still co-occur more (or less) than expected:

- ▶ Unmeasured gradients affect multiple species simultaneously
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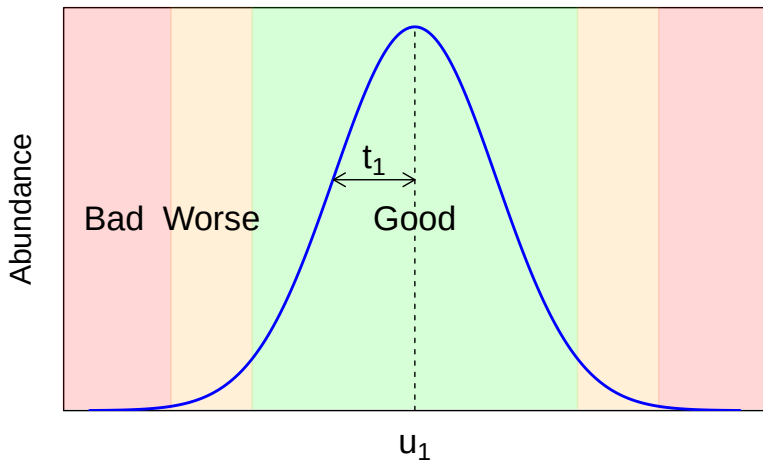
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The latent variables in a JSADM represent unmeasured structure in the ecological process: not distances, not ordination scores, but model parameters with standard errors.

Shelford's law makes community data sparse

Ordination can represent sparse data in only a few dimensions.



What ordination brings

When data are too sparse to estimate species-specific parameters, ordination summarises the main gradients via latent variables:

- ▶ Response-scale dimension reduction: no transformation, no distance metric
- ▶ Uncertainty in ordination scores propagates to all downstream inference
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The cost: rank constraints on species responses induce bias. The number of latent dimensions must be chosen, and the axes require interpretation. Use ordination when the data cannot support a full-rank model.

Prediction matters for community ecology

A model for the data opens up a coherent framework for forecasting:

- ▶ **Species:** occurrence probability or abundance at unvisited sites or future conditions
- ▶ **Communities:** composition, turnover, and beta diversity across gradients
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The same model used for ordination can forecast diversity under climate change.

Summary: classical approach vs. GLLVM

Problem	Classical approach	GLLVM
Count/binary data	$\log(y + 1)$ transformation	Poisson, NB, or binomial distribution
Long gradient (arch)	DCA rescaling (heuristic)	Unimodal response model
Nested / repeated design	None	Random effects
Species correlation	Ignored	JSDM / residual covariance
No environment measured	NMDS	Unconstrained ordination
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Models will be more honest to you than transformations.